caugi: Fast and Flexible Causal Graph Interface for R

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Motivation

- Existing packages often rely on low-level representations (e.g., adjacency matrices) and are not built specifically for causal graphs.
- Causal workflows often bounce between graph objects, adjacency matrices, and package-specific APIs.
- ***caugi* delivers** an expressive, safe, and efficient graph interface for causal inference.

Key Contributions

- **High-performance backend:** Rust implementation for fast, memory-safe graph traversal.
- **Type-safe graph classes:** DAG, PDAG, ADMG, and UG constraints are enforced at the graph level.
- **Expressive syntax:** Compose graphs with concise R formulas that read naturally.
- **Scalable queries:** Efficient ancestry and neighborhood relations on large graphs.

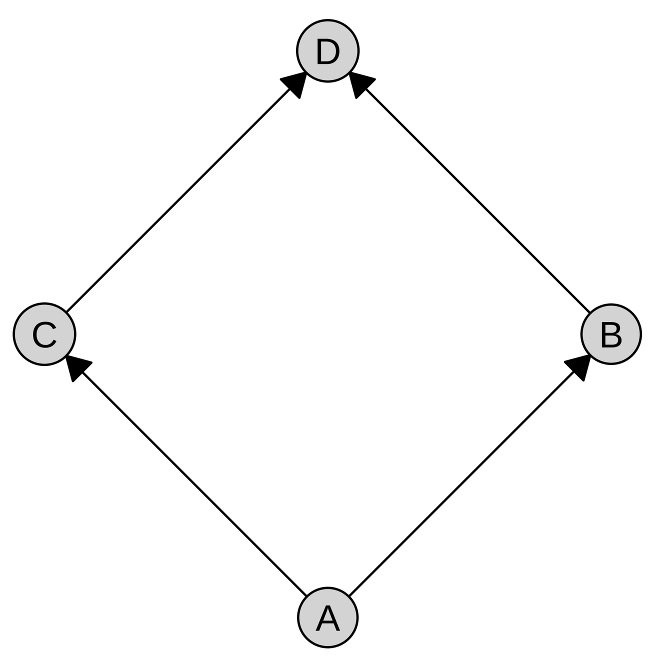
Basic Usage

Available on CRAN:

```r
install.packages("caugi")
library(caugi)
```

Syntax like a picture in your head:

```r
cg <- caugi(
  A %-->% B + C,
  B %-->% D,
  C %-->% D,
  class = "DAG"
)
plot(cg)
```



Querying and Metrics

- **Relational queries**: `parents()`, `ancestors()`, `neighbors()`, etc.
- **Structural queries**: `is_acyclic()`, `is_cpdag()`, etc.
- **Causal queries**: `adjustment_set()`, `d_separated()`, etc.
- **Graph metrics**: `shd()`, `aid()`, etc.

Example queries:

```r
> parents(cg, "D")
[1] "B" "C"

> d_separated(cg, "A", "D", Z = c("B", "D"))
[1] TRUE
```

How it Works

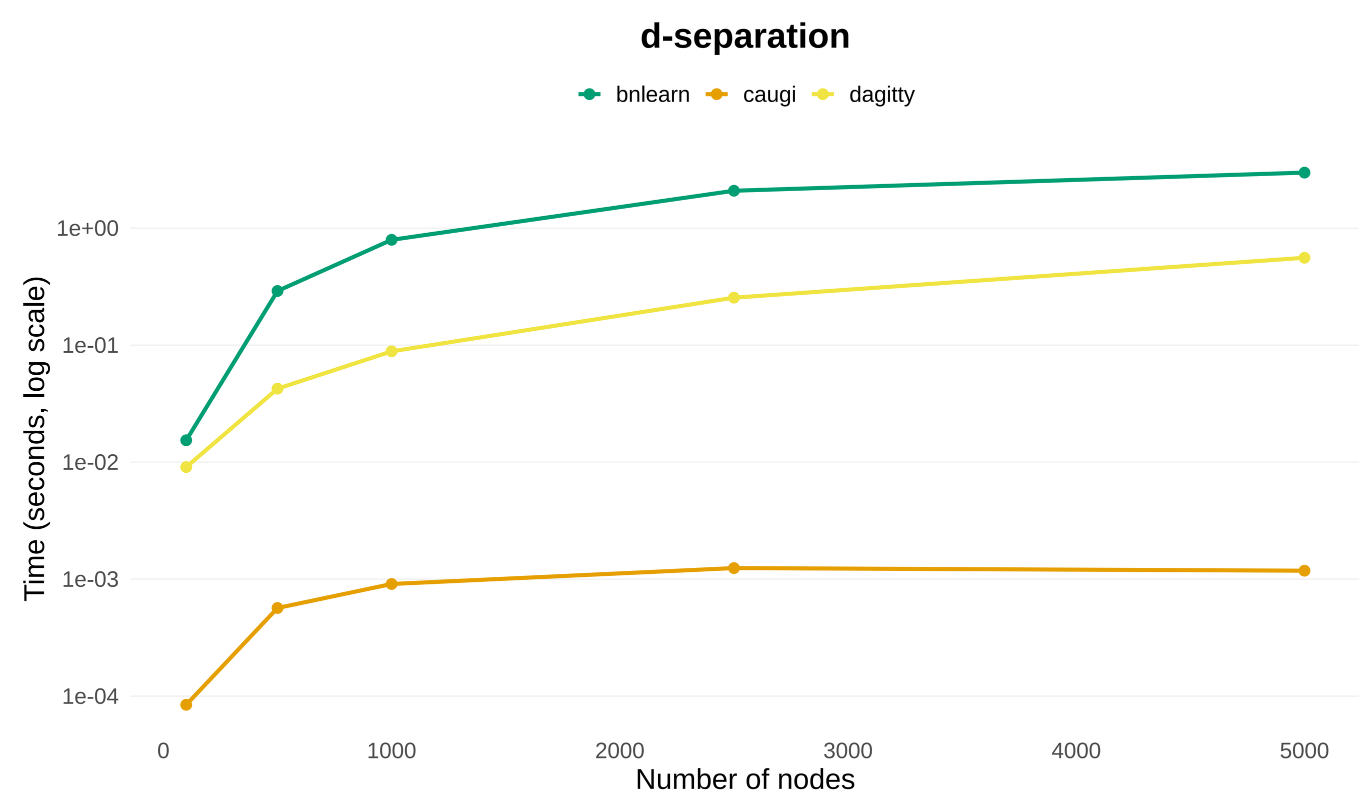
- **Backend:** Rust for memory safety and performance.
- **Storage:** Compressed Sparse Row (CSR)
  - Memory-efficient for sparse graphs
  - Direct slice access → O(1) adjacency lookup
- **Immutable + lazy rebuild:**
  - Rebuilds in O(|V| + |E|) on query after graph modification.
- **Result:** Fast, predictable queries with consistent graph state.

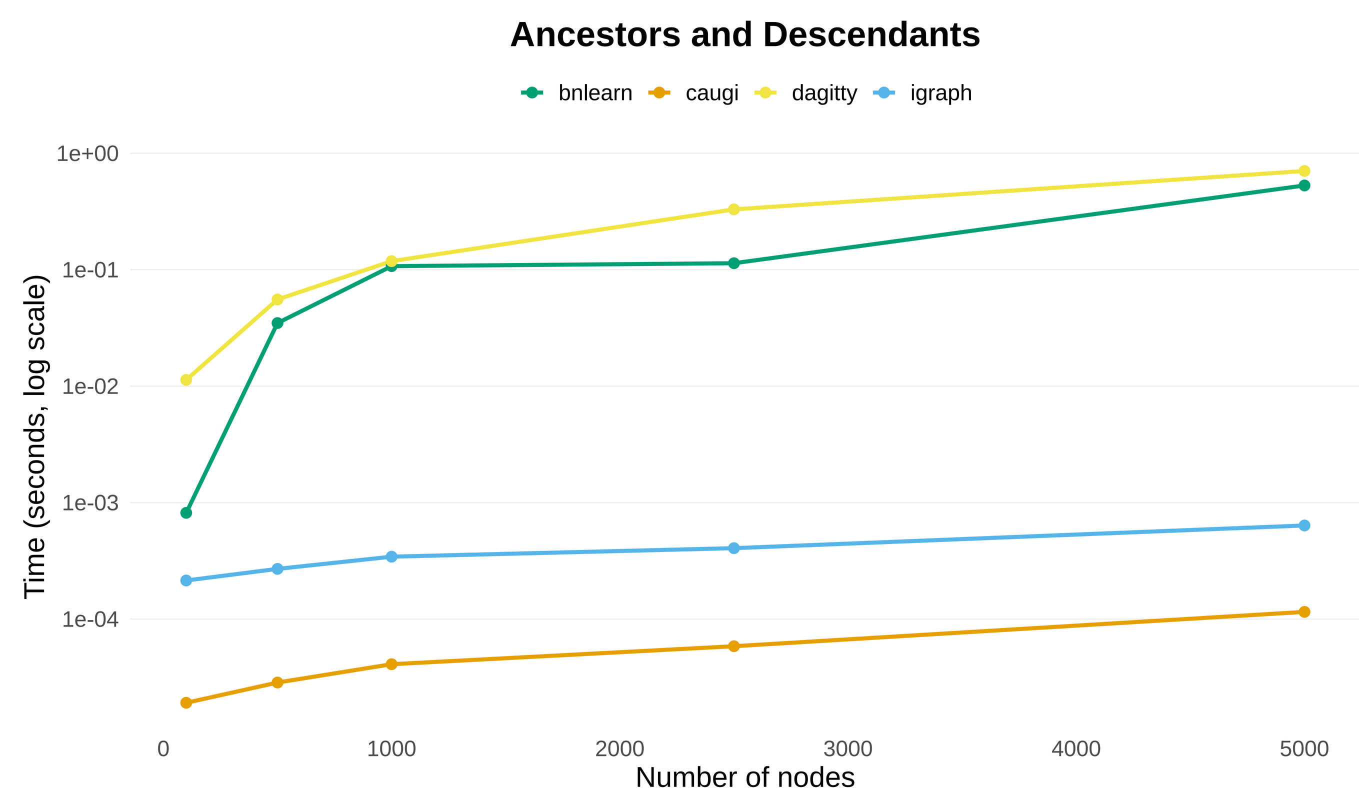
Graph-Class Safe

- Type safety, but for graphs.
- Supports DAGs, PDAGs, ADMGs, UGs, and unknown graphs.
- All graph modifications are checked against class constraints.
  - For example, adding an edge that creates a cycle in a DAG is prevented.
- Keeps graph state reliable and consistent throughout analysis.

Benchmarks

- Benchmarked *caugi* against *bnlearn*, *dagitty*, and *igraph*.
  - We report median runtime.
- *caugi* consistently an order or two faster than alternatives.





Package Documentation

- Pkgdown site: caugi.org

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